

Class 5:

Evaluate the following integrals by changing the order of integration:

1. $\int_0^3 \int_1^{\sqrt{4-y}} (x+y) dx dy$

ans: $\frac{241}{60}$

2. $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx$

ans: $1 - \frac{1}{\sqrt{2}}$
